Adaptive surrogate models for Bayesian inference - Applications to Nuclear Containment Buildings

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Outline

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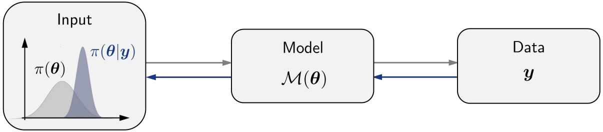
Bayesian inference for inverse problems

- Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space
- Let $\mathcal{M} : \mathcal{D}_{\boldsymbol{\theta}} \subseteq \mathbb{R}^d \longrightarrow \mathbb{R}^n$ a deterministic model with random input parameters $\boldsymbol{\Theta} : \Omega \longrightarrow \mathcal{D}_{\boldsymbol{\theta}}$
- The level of knowledge about $\boldsymbol{\Theta}$ is summarized by a prior density $\pi(\boldsymbol{\theta})$
- Given observed data $\boldsymbol{y} \in \mathbb{R}^n$, the posterior density of input parameters may be derived with Bayes' theorem:

$$\pi(\boldsymbol{\theta}|\boldsymbol{y}) = \frac{\pi(\boldsymbol{\theta})\mathcal{L}(\boldsymbol{\theta};\boldsymbol{y})}{\mathcal{Z}}$$

where $\mathcal{L}(\boldsymbol{\theta};\boldsymbol{y})$ is the likelihood function, and $\mathcal{Z}$ the model evidence:

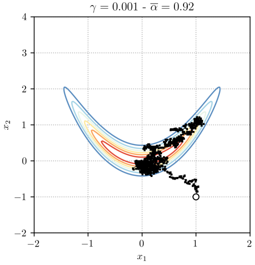
$$\mathcal{Z} = \int_{\mathcal{D}_{\boldsymbol{\theta}}} \mathcal{L}(\boldsymbol{\theta};\boldsymbol{y})\pi(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}$$



Bayesian computations

- Bayesian computations are often based on MCMC sampling techniques (Robert & Casella, 2004; Gamerman & Lopes, 2006)
- MCMC typically requires a large number of likelihood evaluations
- The forward model $\mathcal{M}$ may be costly

▶ Need to accelerate Bayesian computations



Example of a MCMC run

Surrogate modeling for accelerated Bayesian computations

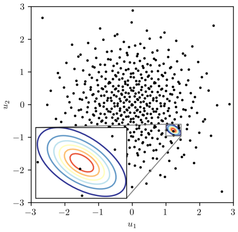
- In order to accelerate Bayesian computations, the forward model $\mathcal{M}$ is usually replaced by a surrogate model $\hat{\mathcal{M}}$:

$$\mathcal{M}(\boldsymbol{\theta}) \approx \hat{\mathcal{M}}(\boldsymbol{\theta})$$

- In this context, prior-based surrogate models are widely used, in order to device a global approximation of the forward model on the prior support

(Marzouk et al., 2007; Sraj et al., 2015)

- However, such approaches may fail if data are strongly informative... (Lu et al., 2015; Yan & Zhou, 2019)
- ... and/or if the forward model is strongly nonlinear on the prior support
- Local accuracy near informative zones may not be guaranteed, even with a large experimental design size



Prior-based experimental design vs. concentrated posterior

Motivations and proposed methodology

- Main goal: construct a surrogate model with an adaptive experimental design, by selecting enrichment points near informative zones

- Proposition of an approach based on the so-called BUS framework *(Bayesian Updating with Structural reliability methods)*, which reformulates Bayesian inference into a Structural Reliability (SR) problem
(Straub & Papaioannou, 2015)

- BUS is used as a basis for devising active-learning schemes inspired from SR methods
(Bichon et al., 2008; Echard et al., 2011; Moustapha et al., 2022)

- Use of Polynomial Chaos Kriging (PCK) surrogates: (Schöbi et al., 2015, 2017)
  - Provide a global approximation on the prior support with Polynomial Chaos Expansions (PCE)...
  - ...as well as local features provided by the Kriging part, for improving local accuracy near informative zones

Outline

The BUS formulation

- Let $c > 0$ be a constant such that $c\mathcal{L}(\boldsymbol{\theta}; \boldsymbol{y}) \leq 1$, for all $\boldsymbol{\theta} \in \mathcal{D}_{\boldsymbol{\theta}}$
- Consider a random variable $P \sim \mathcal{U}([0,1])$ independent from the parameters $\boldsymbol{\Theta}$
- One defines the following failure event in the augmented space $\mathcal{D}_{\boldsymbol{\theta}} \times [0,1]$:

$$\mathcal{F} = \{(\boldsymbol{\theta}, p) \in \mathcal{D}_{\boldsymbol{\theta}} \times [0,1] \mid p \leq c\mathcal{L}(\boldsymbol{\theta}; \boldsymbol{y})\}$$

- The posterior density $\pi(\boldsymbol{\theta}|\boldsymbol{y})$ then rewrites: (Straub & Papaioannou, 2015)

$$\pi(\boldsymbol{\theta}|\boldsymbol{y}) = \frac{1}{\mathbb{P}_{\mathcal{F}}} \int_0^1 \mathbf{1}_{\mathcal{F}}(\boldsymbol{\theta}, p)\mathrm{d}p$$

where $\mathbb{P}_{\mathcal{F}}$ is the failure probability defined by:

$$\mathbb{P}_{\mathcal{F}} = \mathbb{P}((\boldsymbol{\Theta}, P) \in \mathcal{F}) = \int_{\mathcal{D}_{\boldsymbol{\theta}}} \int_0^1 \mathbf{1}_{\mathcal{F}}(\boldsymbol{\theta}, p)\pi(\boldsymbol{\theta})\mathrm{d}p\mathrm{d}\boldsymbol{\theta} = c\mathcal{Z}$$

The BUS formulation

- Sampling from the posterior $\pi(\boldsymbol{\theta}|\boldsymbol{y})$ is tantamount to solving the reliability problem associated to the following Limit State Function (LSF):

$$\mathcal{G}(\boldsymbol{\theta}, p) = p - c\mathcal{L}(\boldsymbol{\theta}; \boldsymbol{y})$$

$$\mathcal{F} = \{(\boldsymbol{\theta}, p) \in \mathcal{D}_{\boldsymbol{\theta}} \times [0, 1] \mid \mathcal{G}(\boldsymbol{\theta}, p) \leq 0\}$$

- ▸ Samples that lie in $\mathcal{F}$ are following the posterior distribution (Straub & Papaioannou, 2015)

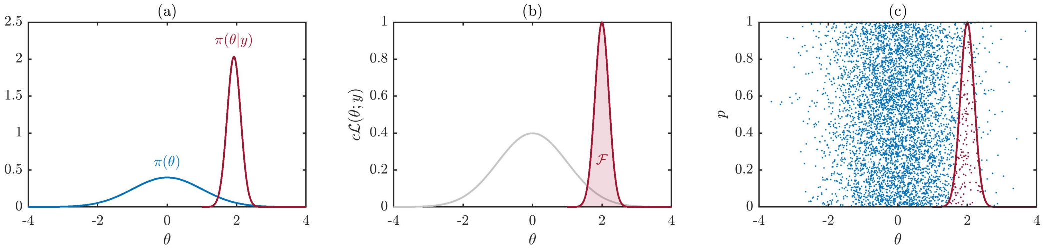


Illustration of the BUS framework: (a) *Bayesian inference*; (b) *Definition of the failure event $\mathcal{F}$*; (c) *Solving the equivalent reliability problem through MCS*

Subset Simulation (SuS) within BUS

- Use of the well-known Subset Simulation (SuS) technique for solving the equivalent SR problem:

$$\mathbb{P}_{\mathcal{F}} = \mathbb{P}\left(\bigcap_{i=0}^{r} \mathcal{F}_i\right) = \prod_{i=1}^{r} \mathbb{P}(\mathcal{F}_i|\mathcal{F}_{i-1})$$

where $\mathcal{F} = \mathcal{F}_r \subset \mathcal{F}_{r-1} \subset \cdots \subset \mathcal{F}_1 \subset \mathcal{F}_0$, with $\mathbb{P}(\mathcal{F}_0) = 1$ (Au & Beck, 2001)

- Definition of the subsets $(\mathcal{F}_i)_{0\leq i\leq r}$:

$$\mathcal{F}_i = \{(\boldsymbol{\theta}, p) \in \mathcal{D}_{\boldsymbol{\theta}} \times [0,1] \mid \mathcal{H}(\boldsymbol{\theta}, p) \leq h_i\}$$

where $0 = h_r < h_{r-1} < \cdots < h_1 < h_0 = +\infty$ are threshold values, and $\mathcal{H}$ is a modified version of the LSF $\mathcal{G}$:

$$\mathcal{H}(\boldsymbol{\theta}, p) = \log(p) - \log(c) - \log(\mathcal{L}(\boldsymbol{\theta}; \boldsymbol{y}))$$

(DiazDelaO et al., 2017; Betz et al., 2018)

- The $(h_i)_{1\leq i\leq r-1}$ are chosen such that the conditional probabilities $(\mathbb{P}(\mathcal{F}_i|\mathcal{F}_{i-1}))_{1\leq i\leq r-1}$ are equal to a prescribed value $\mathbb{P}_t$ in average (with usually $\mathbb{P}_t = 10\%$)

Subset Simulation (SuS) within BUS

- Main steps of SuS procedure:
  - estimation of $\mathbb{P}(\mathcal{F}_1)$ through prior-based MCS
  - estimation of the conditional probabilities $(\mathbb{P}(\mathcal{F}_i|\mathcal{F}_{i-1}))_{2\leq i\leq r}$ through MCMC sampling
- Use of a MCMC algorithm tailored for SuS (Papaioannou et al., 2015)

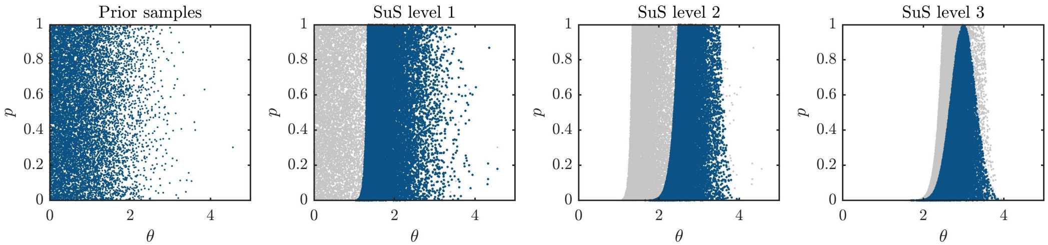


Illustration of SuS procedure, in the case $\pi(\theta) = \mathcal{N}(\theta|0,1)$ and $\mathcal{L}(\theta;y) = \mathcal{N}(\theta|2,0.2)$

The choice of the BUS normalizing constant

- The choice of the normalizing constant c plays a key role in the BUS-SuS procedure
- Optimal choice for c:
$$c_{\text{opt}}^{-1} = \max_{\boldsymbol{\theta} \in \mathcal{D}_{\boldsymbol{\theta}}} \mathcal{L}(\boldsymbol{\theta}; \boldsymbol{y})$$
however, such value is usually not known in advance, and MLE may be intractable
- If $c^{-1} < c_{\text{opt}}^{-1}$, BUS-SuS samples may not follow the posterior distribution...
- ...otherwise, if $c^{-1} > c_{\text{opt}}^{-1}$, the overall performance of BUS-SuS may be altered, since $\mathbb{P}_{\mathcal{F}} \propto c$

▸ Solution adopted: on the fly adaptation of c throughout SuS procedure, by setting c as the maximum likelihood value on samples of the current SuS level

(DiazDelaO et al., 2017; Betz et al., 2018)

Outline

Introduction

The BUS framework

Adaptive surrogate modeling

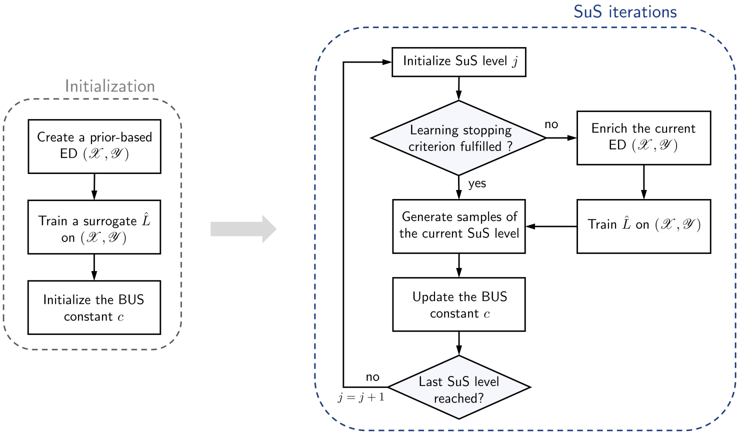
Numerical test cases

Applications to NCB

Conclusions

Adaptive surrogate modeling procedure

- General framework of the proposed approach: (D.R., Baroth, Briffaut, & Dufour, 2021)



PCK-based surrogate modeling

- Use of Polynomial Chaos Kriging (PCK) surrogate models: (Schöbi et al., 2015, 2017)

$$L(\boldsymbol{\theta}) \approx \hat{L}(\boldsymbol{\theta}) = \sum_{\boldsymbol{\alpha} \in \mathcal{A}} a_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{\theta}) + Z(\boldsymbol{\theta})$$

  where $(\psi_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha} \in \mathcal{A}}$ is a truncated basis of multivariate polynomials, orthonormal w.r.t. the prior, and Z a zero-mean stationary Gaussian process

- Polynomial basis $(\psi_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha} \in \mathcal{A}}$ determined trough the LARS-based procedure of (Blatman & Sudret, 2011)

- $(\psi_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha} \in \mathcal{A}}$ may then be entirely embedded in PCK as a Kriging trend (Sequential PCK), or iteratively in order to minimize a cross-validation error (Optimal PCK) (Schöbi et al., 2015)

- PCK surrogates are built with the UQLab software (Marelli & Sudret, 2014)

Learning stopping criterion

- Let $\mu_{\hat{L}}(\boldsymbol{\theta})$ and $\sigma^2_{\hat{L}}(\boldsymbol{\theta})$ be the prediction mean and variance of the PCK surrogate $\hat{L}$ at a point $\boldsymbol{\theta} \in \mathcal{D}_{\boldsymbol{\theta}}$
- Replacing the log-likelihood by its surrogate $\hat{L}(\boldsymbol{\theta})$ yields a surrogate LSF:

$$\hat{\mathcal{H}}(\boldsymbol{\theta}, p) = \log(p) - \log(c) - \hat{L}(\boldsymbol{\theta})$$

- $\hat{\mathcal{H}}$ is also a Gaussian process, with $\hat{\mathcal{H}}(\boldsymbol{\theta}, p) \sim \mathcal{N}\left(\log(p) - \log(c) - \mu_{\hat{L}}(\boldsymbol{\theta}), \sigma^2_{\hat{L}}(\boldsymbol{\theta})\right)$
- Use of the well-known U learning function based on surrogate LSF predictions: (Echard et al., 2011)

$$U(\boldsymbol{\theta}, p) = \frac{|\mu_{\hat{\mathcal{H}}}(\boldsymbol{\theta}, p)|}{\sigma_{\hat{\mathcal{H}}}(\boldsymbol{\theta}, p)}$$

- Use of a standard stopping criterion: $\min U > \epsilon_U$ (with usually $\epsilon_U = 2$) (Echard et al., 2011)

Point enrichment procedure

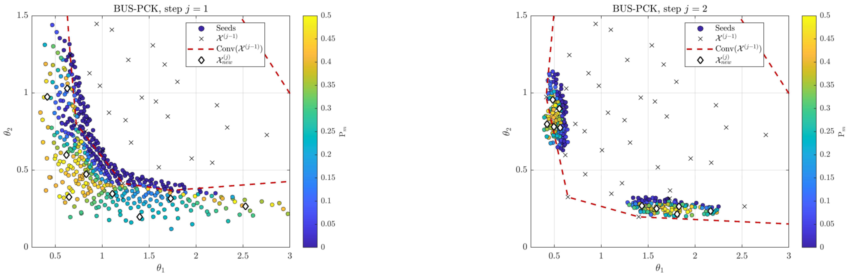
At each SuS level, before the beginning of MCMC sampling:

- The Markov chain seeds are ranked according to their misclassification probability: (Bect et al., 2011)

$$\mathbb{P}_m(\boldsymbol{\theta}, p) = \Phi\left(-U(\boldsymbol{\theta}, p)\right)$$

  where Φ is the standard normal CDF

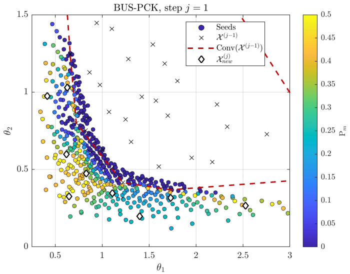
- If current ED is enriched by selecting the point which maximizes $\mathbb{P}_m$, or by using weighted k-means clustering in the case of multiple point enrichment

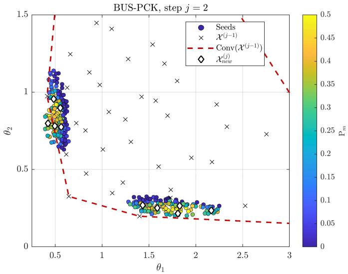


On the fly updating of the BUS scaling constant

At the end of each SuS level:

- The scaling constant c is updated by taking the maximum likelihood value on the current SuS samples (DiazDelaO et al., 2017; Betz et al., 2018)
- Samples which are too far away from the current ED may corrupt the estimation of c
  - Proposed solution: update c based on samples which lie in the convex hull of the current ED (Barber et al., 1996)





Outline

Numerical tests cases: general settings

- Input parameters of the proposed approach: initial ED size K_0, number of enrichment points per subset K_t
- Reference results produced with the BUS-SuS method
- Error measure on univariate posterior marginals: averaged Jensen-Shannon-Divergence (aJSD)

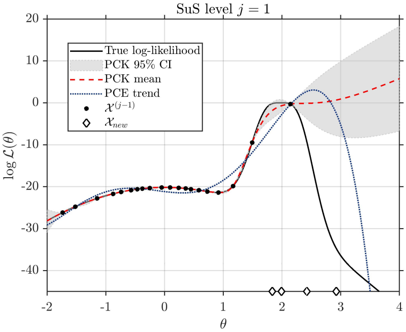
$$\overline{\Delta}_{\mathrm{JS}}(\pi||\hat{\pi}) = \frac{1}{d}\sum_{i=1}^{d} \Delta_{\mathrm{JS}}(\pi_i(\cdot|\boldsymbol{y}), \hat{\pi}_i(\cdot|\boldsymbol{y}))$$

(Wagner et al., 2021)

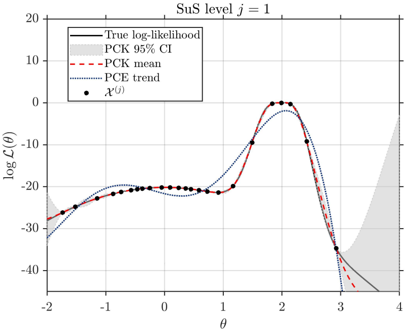
where d is the input dimension, $\pi_i(\cdot|\boldsymbol{y}), \hat{\pi}_i(\cdot|\boldsymbol{y})$ are the reference and surrogate-based posterior marginals, and Δ_{JS} is the Jensen-Shannon Divergence (JSD) (Lin, 1991)

Example 1: 1D analytical function

- Example of a 1D analytical function: $\mathcal{M}(\theta) = 1 + \cos\left(\frac{\theta}{2}\right) + 3\exp(-4(\theta-2)^2)$
- Model setup: $\Theta \sim \mathcal{N}(0,1)$ and $\mathcal{L}(\theta; y) = \mathcal{N}(\mathcal{M}(\theta)|y, \sigma^2)$, with $y = \mathcal{M}(2)$ and $\sigma = 0.4$

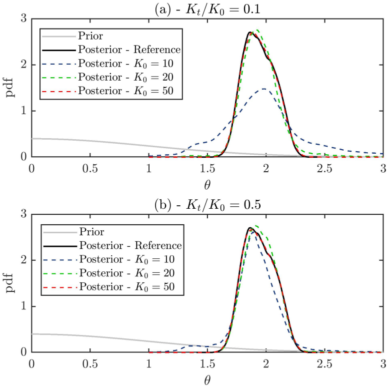


Log-likelihood at SuS level 1, before point enrichment

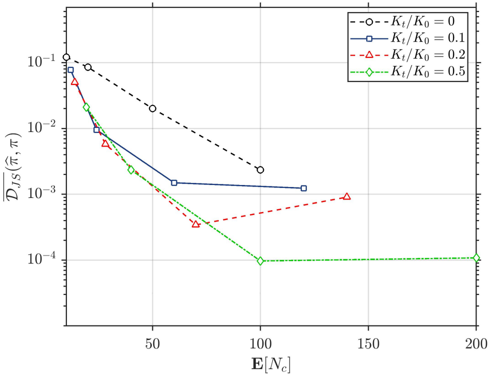


Log-likelihood at SuS level 1, after point enrichment

Example 1: 1D analytical function



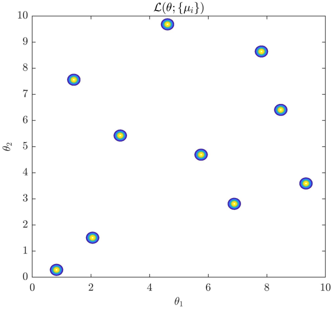
Prior and posterior densities



aJSD vs. average total number of model calls $\mathbb{E}[N_c]$

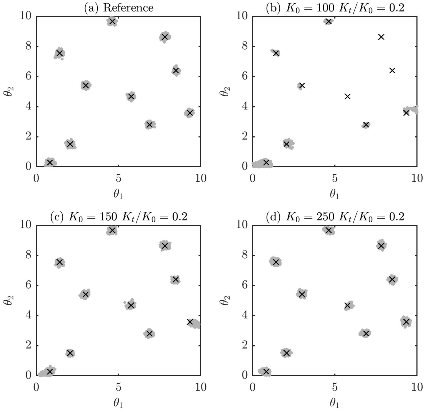
Example 2: 2D multi-modal posterior

- Example of a multi-modal posterior distribution, with 10 distinct modes (Beck & Zuev, 2013)
- Prior: $\boldsymbol{\Theta} \sim \mathcal{U}([0,10]^2)$; Data: $\boldsymbol{y} = \{\boldsymbol{\mu}_i\}_{1 \leq i \leq m}$, with $\boldsymbol{\mu}_i \sim \mathcal{U}([0,10]^2)$ and $m = 10$
- Likelihood: $\mathcal{L}(\boldsymbol{\theta}; \boldsymbol{y}) = \frac{1}{m} \sum_{i=1}^{m} \mathcal{N}(\boldsymbol{\theta} | \boldsymbol{\mu}_i, \sigma^2)$

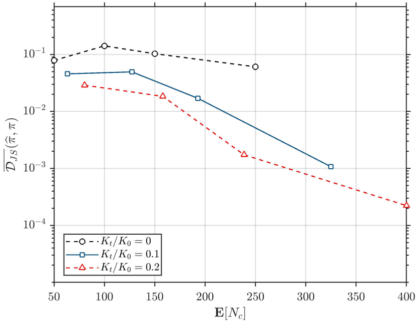


Contours of the likelihood $\mathcal{L}(\boldsymbol{\theta}; \boldsymbol{y})$

Example 2: 2D multi-modal posterior



Posterior samples



aJSD vs. average total number of model calls $\mathbb{E}[N_c]$

Example 3: diffusion problem

- Example of a 1D diffusion problem with a random diffusivity field (Marzouk et al., 2007; Straub et al., 2016)

- Diffusion equation, with N sources with locations $(l_i)_{1\leq i\leq N}$, strengths $(s_i)_{1\leq i\leq N}$ and widths $(\sigma_i)_{1\leq i\leq N}$:

$$\nabla\left(a(x)\nabla u(x)\right)+\sum_{i=1}^{N}\frac{s_i}{\sqrt{2\pi}\sigma_i}\exp\left(-\frac{(l_i-x)^2}{2\sigma_i^2}\right)=0$$

  where $a(x)$ is the diffusivity field, and $u(x)$ the unknown field

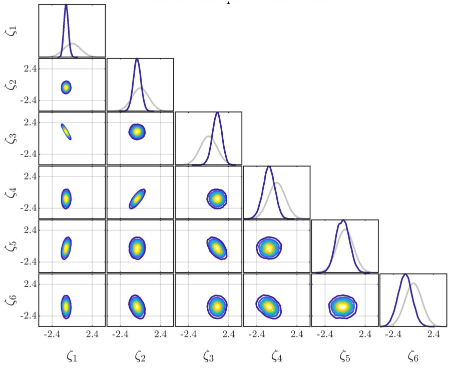
- Prior log-diffusivity field: Gaussian stationary random field with exponential correlation kernel

- Discretization of $\log a(x)$ with a truncated Karhunen-Loève expansion, with $m=10$ eigenmodes:

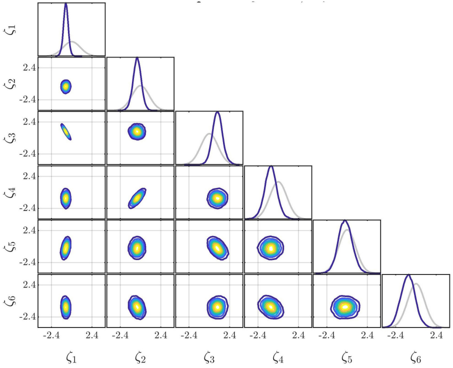
$$\log a(x)\approx\mu_{\log a}+\sum_{i=1}^{m}\sqrt{\lambda_i}\theta_i\phi_i(x)$$

  - Inference of $\boldsymbol{\theta}=(\theta_i)_{1\leq i\leq m}$ from measurements of the field $u(x)$ corresponding to a realization of $a(x)$

Example 3: diffusion problem

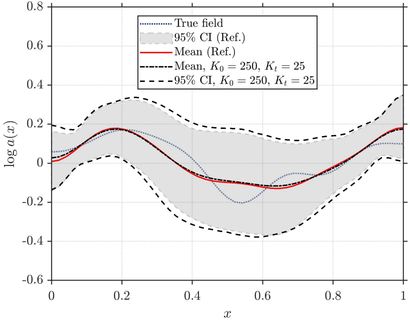


Reference posterior

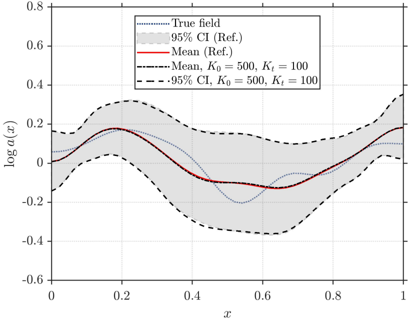


Surrogate-based posterior, $K_0 = 250$, $K_t = 25$

Example 3: diffusion problem

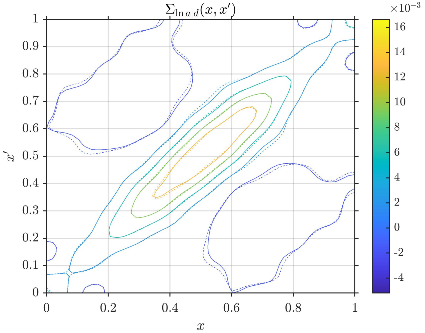


Posterior log-diffusivity field, $K_0 = 250$, $K_t = 25$

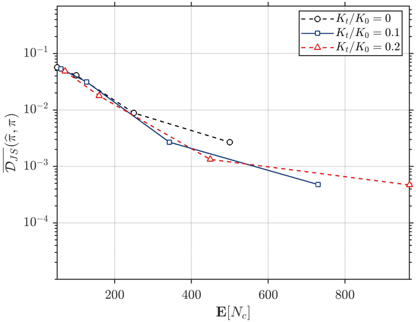


Posterior log-diffusivity field, $K_0 = 500$, $K_t = 50$

Example 3: diffusion problem



Posterior covariance kernel, $K_0 = 500$, $K_t = 100$ *(Continuous lines: reference; dashed lines: surrogate-based)*



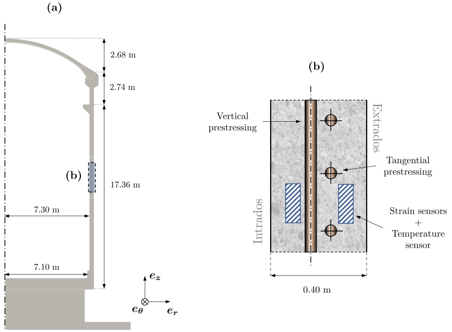
aJSD vs. average total number of model calls $\mathbb{E}[N_c]$

Outline

Case study: the VeRCoRs mock-up

- VeRCoRs mock-up: 1:3 scale double-walled NCB mock-up (Charpin et al., 2021)
- Well instrumented structure involving a large amount of monitoring data
- Structural (and local) leakage rate measured through periodic pressurization tests

Maquette VeRCoRs, EDF-LAB Les Renardières



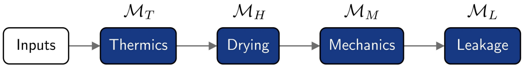
(a) 2D view of VeRCoRs mock-up; (b) Focus on a standard zone

Modeling the physical behavior of NCB

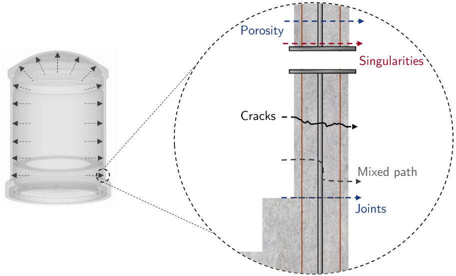
- Modeling of the spatio-temporal evolution of thermal, hydrous, mechanical and transfert quantities of interest (*e.g.* water saturation, strains, leakage rate)

- Use of a weakly-coupled thermo-hydro-mechanical & leakage (THML) model: (D.R., Bouhjiti, et al., 2021)

$$\mathcal{M} = \mathcal{M}_L \circ \mathcal{M}_M \circ \mathcal{M}_H \circ \mathcal{M}_T$$



Weakly-coupled THML modeling



Phenomenology of leaks through the inner wall during a pressurization test

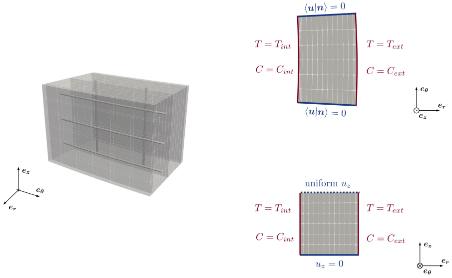
Finite element model and input parameters

- Use of Representative Structural Volumes (RSV) for modeling the behavior of the mock-up

(D.R., Baroth, Briffaut, Dufour, Monteil, et al., 2021)

Identification of 9 influent model parameters:

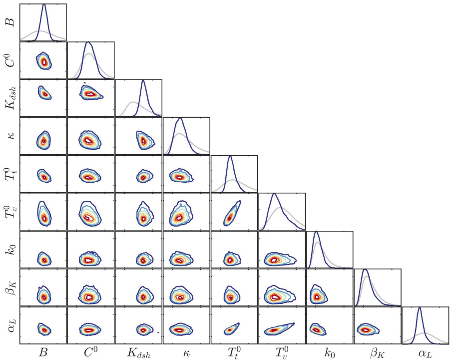
- 2 thermo-hydrous parameters: (B, C^0)
- 4 mechanical parameters: $(K_{dsh}, \kappa, T_t^0, T_v^0)$
- 3 leakage parameters: (k_0, β_K, α_L)



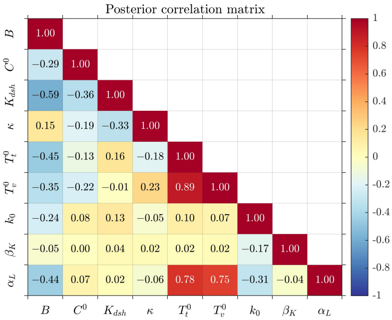
Finite element mesh of VeRCoRs standard zone RSV & boundary conditions

Bayesian updating of model parameters

- Bayesian updating of model parameters using strain and leakage monitoring data



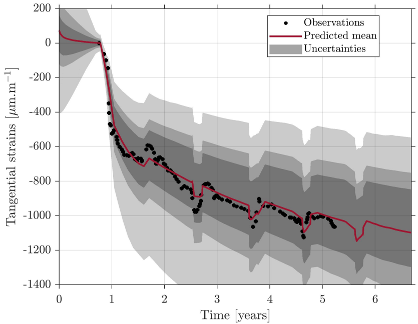
Posterior univariate and bivariate marginal densities



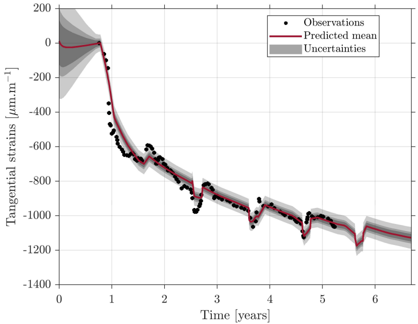
Posterior correlation matrix of model parameters

Probabilistic delayed strains predictions

- Probabilistic predictions of tangential delayed strains of VeRCoRs standard zone



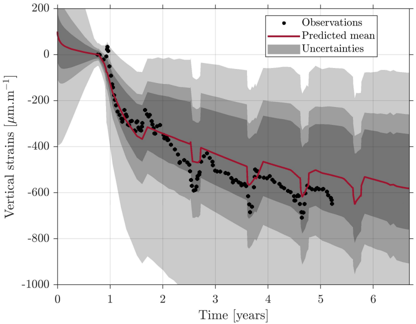
Prior tangential strains of VeRCoRs standard zone *(Mean & 50, 70, 95% CI)*



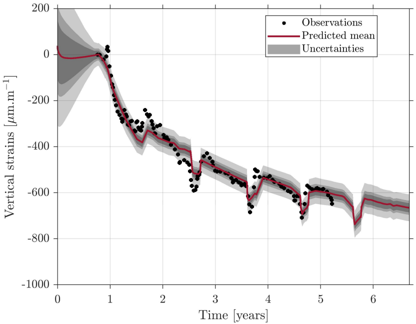
Posterior tangential strains of VeRCoRs standard zone *(Mean & 50, 70, 95% CI)*

Probabilistic delayed strains predictions

- Probabilistic predictions of vertical delayed strains of VeRCoRs standard zone



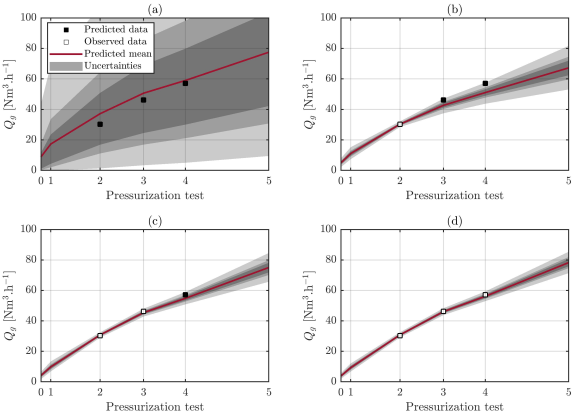
Prior vertical strains of VeRCoRs standard zone *(Mean & 50, 70, 95% CI)*



Posterior vertical strains of VeRCoRs standard zone *(Mean & 50, 70, 95% CI)*

Probabilistic global leakage predictions

- Probabilistic predictions of the global leakage rate of VeRCoRs mock-up



Predictions of the VeRCoRs global leakage rate *(Mean & 50, 70, 95% CI)*: (a) prior predictions; (b) after Test 1; (c) after Test 2; (d) after Test 3

Outline

Conclusions

- Development of a Bayesian updating approach combining the BUS framework with adaptive surrogate models
- The proposed method is particularly efficient for sampling posteriors with "localized" behaviors (*e.g.* concentrated posteriors, multi-modalities)
- Considerable reduction of the computational cost required by Bayesian sampling techniques ($\sim 10^2 - 10^3$ for surrogate-based calculations)
- The proposed approach may be extended in order to update probabilities of failure in a Bayesian framework (possible application to real NCB with regulatory leakage thresholds) (Straub et al., 2016)
- Improvement perspectives related to the choice of the learning stopping criterion, and to the handling of high dimensionalities (> 20)

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